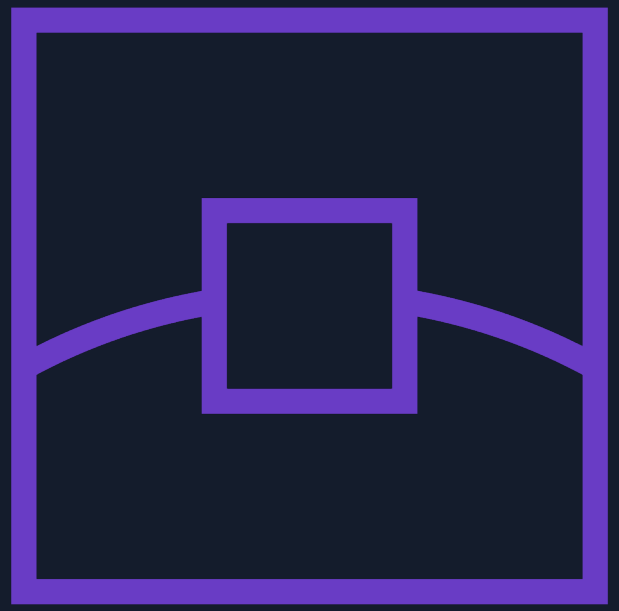
Amazon Web Services: CloudFront

Content delivery networks (CDNs)

- Designed to support scalability, to increase reliability and performance, and to provide better security
- Receives the content from an **origin** server, then replicates it to its **edge** cache servers
  - The content is delivered to an end-user from the “closest” edge server
- Protocols used by CDNs: Network Element Control Protocol (NECP), Web Cache Coordination Protocol (WCCP), SOCKS, Cache Array Routing Protocol (CARP), Internet Cache Protocol (ICP), Hypertext Caching Protocol (HTCP), and Cache Digest

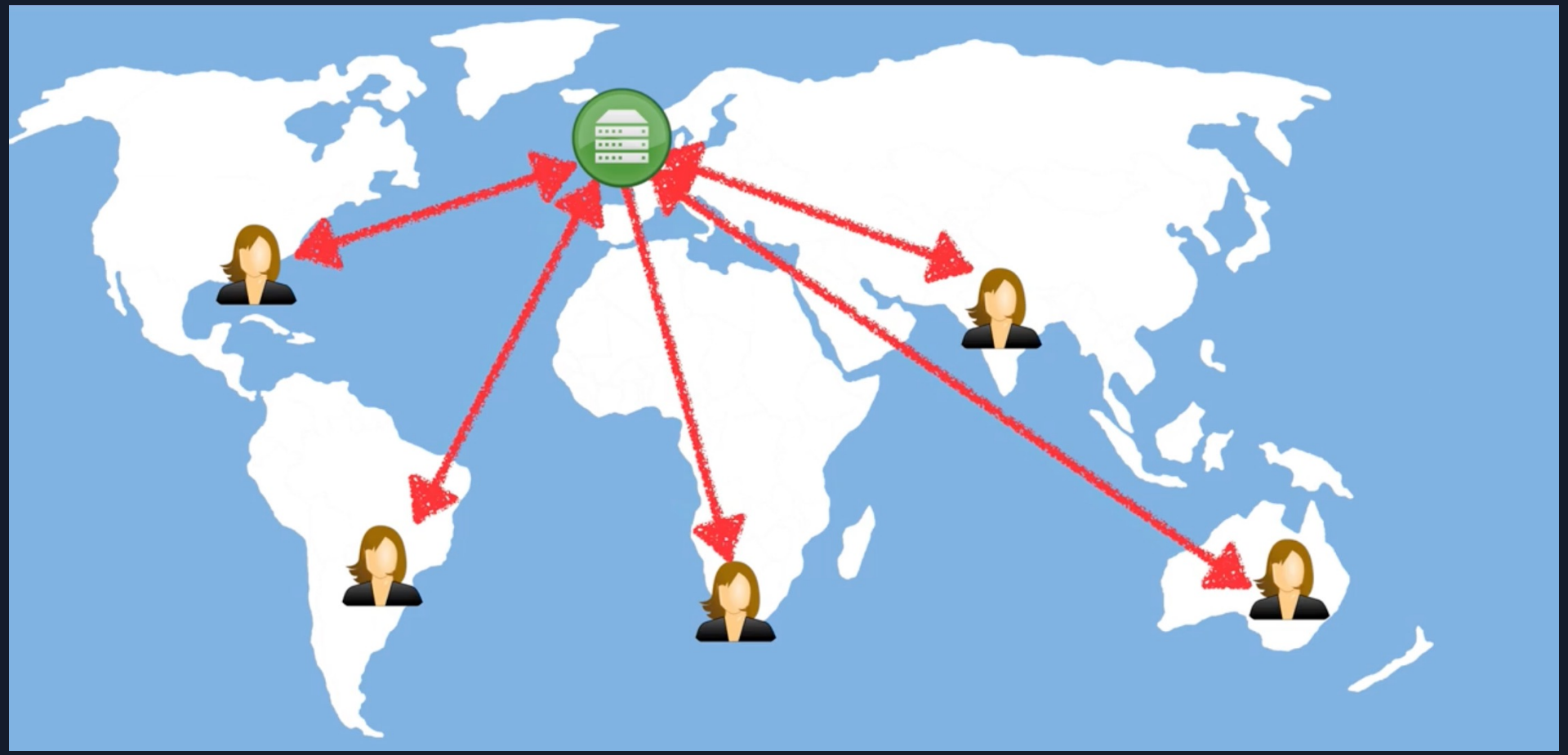
Key Terminology

- Edge Location
  - Location where content is going to be cached
  - Separate from an AWS region or availability zone
  - Read from and write to Edge Location
- Origin
  - Origin of all the files that the CDN will distribute
  - S3 bucket, an EC2 instance, an elastic load balancer, ...
- Distribution
  - Name given to the CDN which consists of a collection of edge locations
  - Web distribution for websites



Edge location

Scenario: CDN != Enabled

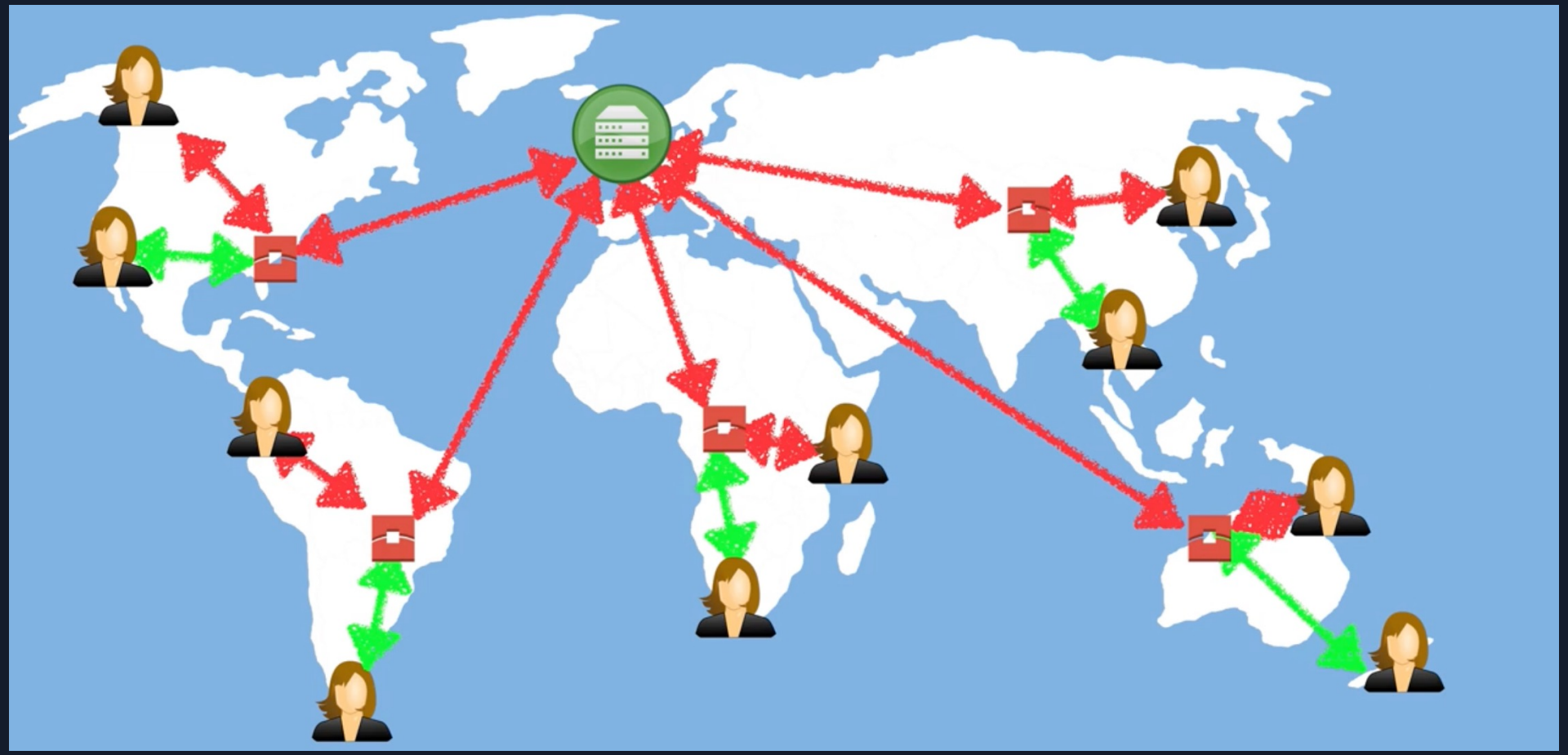


What is CloudFront?

- CloudFront is Amazon's Content Delivery Network (CDN)
- Used to deliver your entire website content using a global network of Edge Locations
- Request for content is automatically routed to the nearest Edge Location
- Helps reduce latency
- Provides higher transfer speeds using those edge locations

Amazon CloudFront

Scenario: CDN == Enabled



Important CloudFront Setting

- Security
  - Defaults to HTTPS connections with the ability to add custom SSL certificate
- Global Distribution
  - You can’t pick specific countries – only general areas of the globe
- Endpoint Support
  - Can be used to front AWS endpoints along with non-AWS applications
- Expiring Content
  - Can force an expiration of content from the cache if you can’t wait for the TTL

Demo: CloudFront

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